3(a)	A longitudinal wave is a wave in which the oscillation of the particles is parallel to the propagation of the energy of the wave. *Common mistakes:* *Some students not clear in their explanation. It is wrong to state particles travel or propagate instead of oscillate or vibrate. Some students did not state clearly that it is the particles that are oscillating.*	[1] for oscillations of particles
3(b)(i)	$\text{phase difference} = \dfrac{\text{path difference}}{\lambda} \times 2\pi$ $\dfrac{3}{2}\pi = \dfrac{6.38 - 5.25}{\lambda} \times 2\pi$ $\lambda = 1.51\text{ m}$	[1] for correct equation and substitution [1] for correct answer

3(b)(ii)	$v = f\lambda$ $= (220)(1.51)$ $= 331\ \text{m s}^{-1}$	[1] for correct equation and substitution. [1] for correct answer
3(c)	$I = kA^2$ $3I = kA'^2$ $3kA^2 = kA'^2$ $A' = \sqrt{3}A$ resultant amplitude $= \sqrt{3}A + A = \left(\sqrt{3}+1\right)A$ resultant intensity $= k\left(\sqrt{3}+1\right)^2 A^2$ $= 7.46kA^2 = 7.46I$ *<u>Common mistakes:</u>* *Some students thought they could just sum up the individual intensities. Some thought that O is a point of destructive interference*.	[1] for amplitude of S_2 [1] for resultant amplitude [1] for resultant intensity
4(a)(i)	Path difference = 2 – 0.020	[1] for correct